NET-UART-9120 User Manual

Version: V1A

http://wch.cn

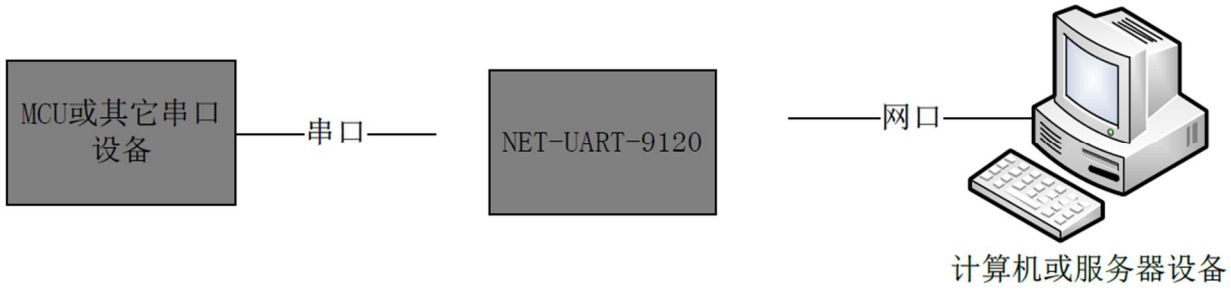
1. Overview

NET-UART-9120 is a network serial port pass-through module, which can realize two-way transparent transmission of network data and serial data, with 4 working modes of TCP CLIENT, TCP SERVER, UDP SERVER, UDP CLIENT, and the maximum baud rate of serial port can be supported to 921600bps, which can be easily configured by the host computer software, convenient and fast.

2. Pin description

RSTI	Input	External reset input, low level valid
RESET	Input	Restore factory settings, chip power-up detection, low level valid
RXD	Input	Serial data input for asynchronous serial port
TXD	Output	Serial data output for asynchronous serial port
RS485DIR	Output	RS485 serial port send/receive direction control pins
GND	Power	Common ground terminal
3V3c	Power	3.3V power output
3V3	Power	5V power input
CFG0	Input	Serial configuration mode setting pin, built-in pull-up, when a low level is detected, enter serial configuration mode, high level exit configuration mode
TCPCS	Output	TCP client mode, connection status indication, low valid

3. Application block diagram and usage flow



The use process is as follows:

(1) Power up the module and connect the module to the computer via a network cable (or connect it to the same LAN as the computer).

(2) According to the actual application mode (working mode, serial port baud rate, etc.), the module configuration can be changed through the host software, as detailed below.

(3) After the configuration is completed, the serial port can be connected in the actual application environment to realize the two-way transparent transmission between the serial port and the network.

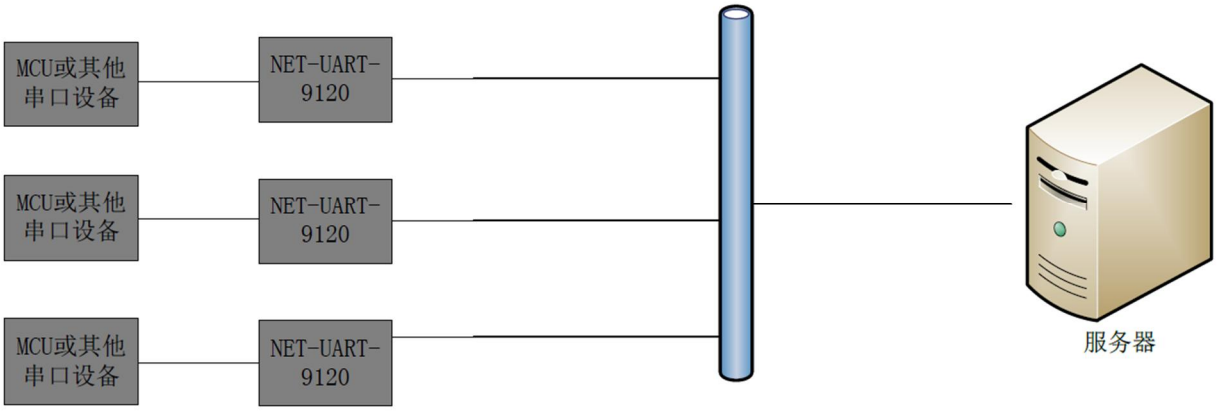
4. Module configuration and operating mode description

The module parameter configuration is divided into two parts: basic setting and port configuration. The basic configuration mainly includes, device name, network parameters, serial port negotiation authentication function, which is off by default and enters serial port setting mode through hardware CFG0 pin by default, and can enter serial port setting mode through serial port negotiation after opening; CH9120 supports DHCP and manual two ways to set network basic parameters.

4.1 TCP CLIENT mode

In TCP CLIENT mode, after the module is powered on, it will actively connect to the TCP SERVER side, and after the connection is established, it can realize the transparent transmission of network data and serial data in both directions. In this mode, the IP of TCP SERVER needs to be visible to the module, which means that the IP of the module can be directly pinged through the server IP (see Section IV for details). TCP client application model is as follows, suitable for field data collection, upload server mode.

CH9120 application block



4.2 TCP SERVER mode

In TCP SERVER mode, after the module is powered on, it will listen to the local port to see if there is a client requesting a connection, and after the connection is established, it can realize the transparent transmission of network data and serial data in both directions. In this mode, the IP of TCP CLIENT needs to be visible to the module, which means that the module IP can be directly pinged through the client IP (see Section IV for details). The network parameters that need to be configured for the module are: operating mode, module IP, subnet mask, default gateway, and module port. In this mode, only one TCP client connection can be supported.

4.3 UDP CLIENT mode

In UDP CLIENT mode, the data sent to the local port (from the destination IP and port) will be transparently forwarded to the module serial port after the module is powered on. Similarly, the data sent to the module serial port will also be forwarded to the set destination IP and port by UDP. In this mode, the module needs to configure the following network parameters: operating mode, module IP, subnet mask, default gateway, module port, destination IP, destination port.

4.4 UDP SERVER mode

In UDP SERVER mode, all data sent to the local IP and port is received and forwarded to the serial port, and data sent to the module's serial port is also forwarded to the IP and port of the UDP with which it communicates via UDP. In this mode, the module needs to be configured with the following network parameters: operating mode, module

IP, subnet mask, default gateway, and module port.

4.5 Configuration software usage instructions and examples

The complete process of using the configuration software is as follows:

(1) Click "Search Module", the module list will show the modules in the subnet;

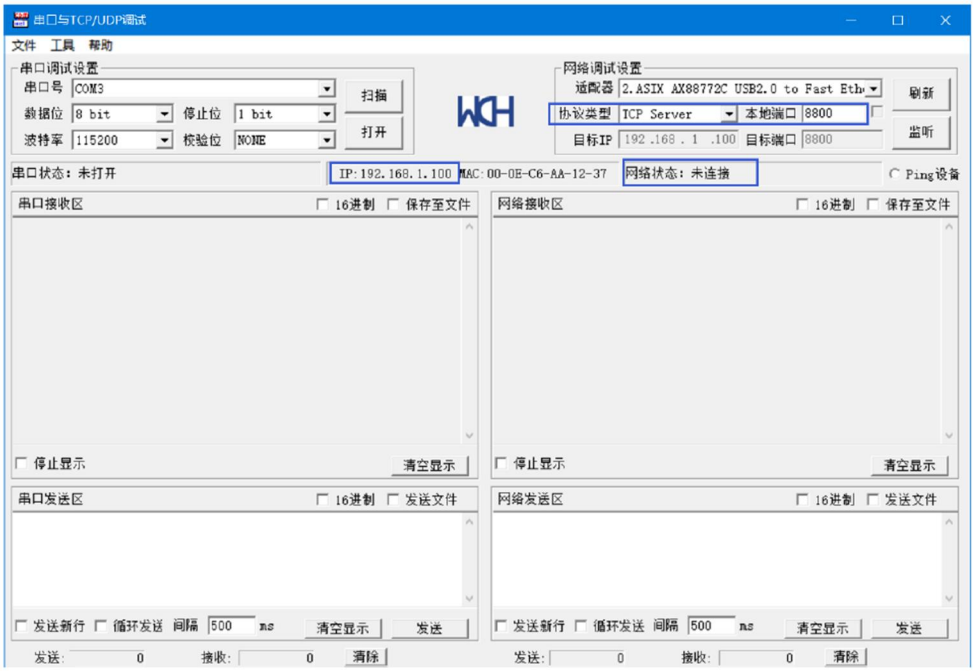
(2) Select the module in the module list, modify the parameters on the left side, and then click "Configure Module Parameters";

(3) After the configuration is finished, the module will restart, wait for a while, click "Search Module", find the module and check the configuration result.

Example: The CH9120 works in TCP client mode, and a TCP server is set up on the computer to communicate via TCP.

(1) Power up the module and connect it directly to the computer via a network cable (or connect the module and the computer to the same LAN).

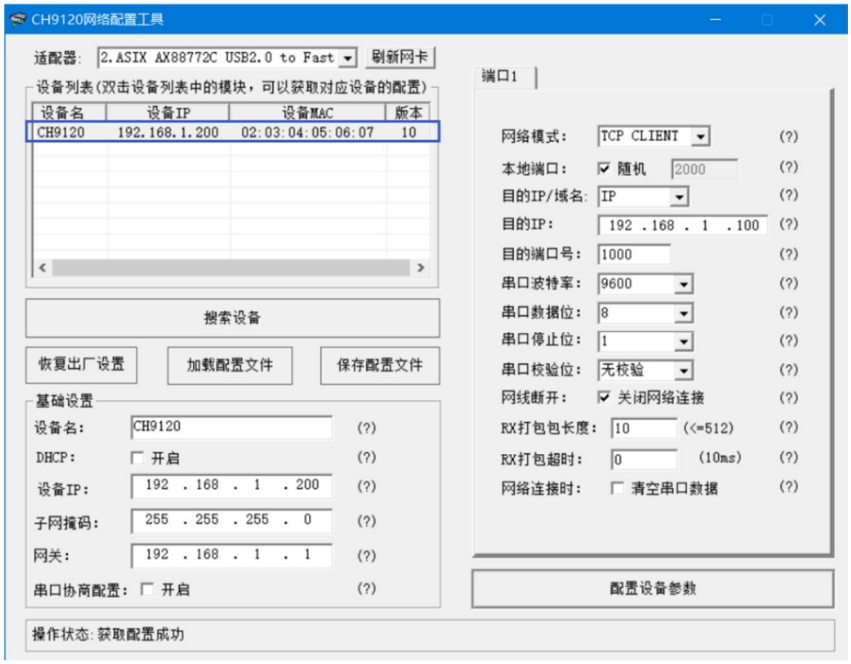
(2) Run the serial port and TCP/UDP debugging tool on the computer side to create a TCP server. Set up as follows: Where the protocol type is TCP SERVER, the local IP is 192.168.1.100, the port is 1000, and the current connection status: not connected.



(3) Run NetModuleConfig.exe to set the module parameters as follows:
Search for the module on a stand-alone basis and find the device.



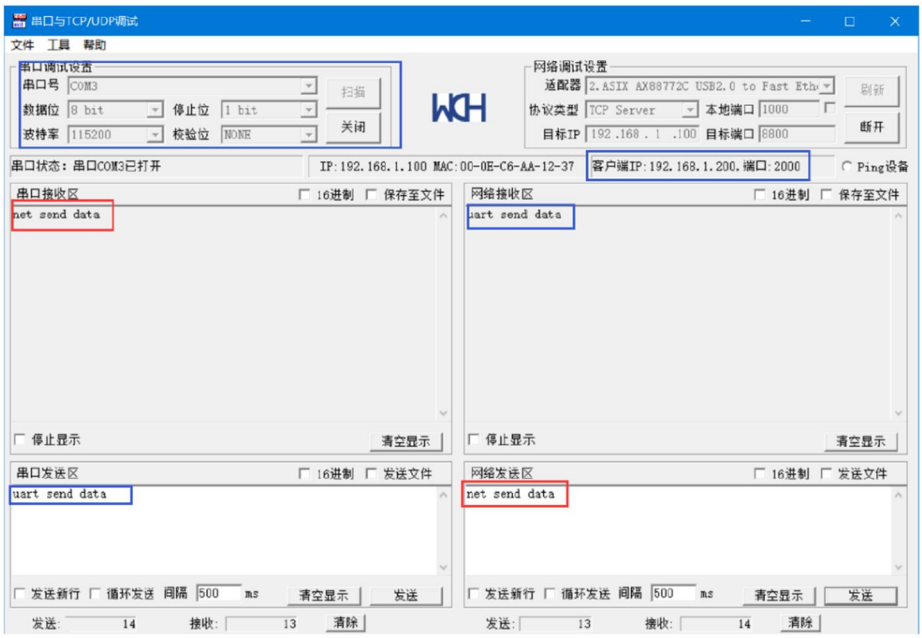
Double click on the searched device to get the parameters.



Modify the module parameters according to the actual demand, set the network method to TCP CLIENT, destination IP, destination port and TCP SERVER's IP and port: 192.168.1.100:1000. modify the baud rate and other parameters according to the demand. Then click Configure Module



(4) At this time, the serial debugging software on the computer side shows that the CH9120 module client is connected, select the serial parameters, and open the serial port for data sending and receiving test.



5. Frequently asked questions and precautions

(1) The upper computer software cannot search the module?

(a) Check whether the module is directly connected to the PC or in the same subnet. For example, when the subnet mask is 255.255.255.0, 192.168.1.1 and 192.168.1.2 are in the same subnet, while 192.168.1.1 and 192.168.2.1 are in different subnets.

(b) check whether the network card selection is correct. This is mainly for multi-NIC PC, such as notebooks generally have a wired NIC and a wireless NIC, when configuring the module, you need to select the wired

NIC instead of the wireless NIC.

(2) The module is working in TCP CLIENT mode and cannot establish a connection to the server?

(a) Check whether the destination port and IP of the module are consistent with the server port and IP.

(b) Check whether the server side is able to ping through the client, if not, check whether the RJ45 (network port) is abnormal, and check whether the two are in the same LAN.

(c) Check whether the firewall is open, it is recommended to close the firewall in this mode to prevent the firewall from blocking the module request.

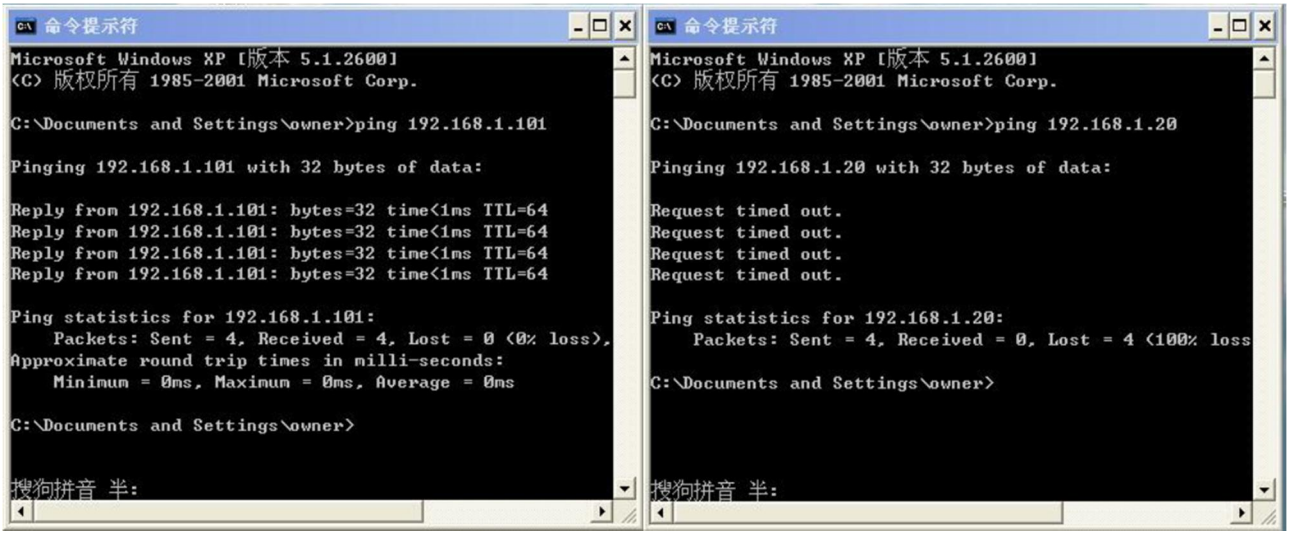
(3) Serial data sending and receiving abnormalities (unable to send and receive data or data error)?

(a) TTL version of the module serial level 3.3V, can be directly connected to the microcontroller serial port, do not directly connected to the 232 level serial port.

(b) (b) The serial port of the module and the microcontroller need to be crossed when connecting, i.e. TXD to RXD, RXD to TXD.

(4) How do I run the PING program, or determine if the PING was successful?

Take WINDOWS system as an example, click "Start"→"All Programs"→"Accessories"→"Command Prompt", then type "PING xxx.xxx.xxx.xxx", where xxx.xxx.xxx.xxx is the module IP. The result is shown in the following figure:



(a)PING Success (b) PING Failure